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Subject: Vsec

Assignment No. 07

1. Verify Docker Installation

Step 1.1 — Check Docker version

docker --version

Expected output:

- Something like: Docker version 26.x.x

Step 1.2 — Check Docker Compose

docker compose version

If this fails:

- Install plugin:

sudo apt install docker-compose-plugin

---

2. Create Project Structure

Step 2.1 — Create root project folder

mkdir ai_web_app

Step 2.2 — Enter the directory

cd ai_web_app

Step 2.3 — Create subfolders

mkdir ai_model frontend database

Step 2.4 — Verify structure

ls

Expected output:

ai_model frontend database

---

3. Build AI Model Service (Flask)

Step 3.1 — Enter AI model directory

cd ai_model

Step 3.2 — Create Flask app

nano app.py

Paste:

```
from flask import Flask, jsonify
import numpy as np

app = Flask(__name__)

@app.route("/")
def predict():
    result = np.mean([10, 20, 30, 40])
    return jsonify({"prediction": result})

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Save:

- Press CTRL + X
- Press Y
- Press ENTER

Step 3.3 — Create requirements file

nano requirements.txt

Paste:

```
flask
numpy
```

Save and exit.

Step 3.4 — Create Dockerfile

nano Dockerfile

Paste:

```
FROM python:3.10-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py .
CMD ["python", "app.py"]
```

Save and exit.

Step 3.5 — Verify files

ls

Expected:

```
app.py  requirements.txt  Dockerfile
```

---

4. Build Frontend (Static HTML)

Step 4.1 — Go back to root

cd ..

Step 4.2 — Enter frontend

cd frontend

Step 4.3 — Create HTML file

nano index.html

Paste:

```
<!DOCTYPE html>
<html>
<head>
<title>AI Web App</title>
</head>
```

```
<body>

<h1>Web-Based AI Application</h1>
<p>AI Model Output will be displayed here</p>

</body>
</html>
```

Step 4.4 — Create Dockerfile

nano Dockerfile

Paste:

```
FROM nginx:alpine
COPY index.html /usr/share/nginx/html/index.html
```

Step 4.5 — Verify files

ls

Expected:

```
index.html  Dockerfile
```

---

5. Docker Compose Setup

Step 5.1 — Go to root folder

cd ..

Step 5.2 — Create compose file

nano docker-compose.yml

Paste:

```
version: "3.8"

services:

  ai_model:
```

```
    build: ./ai_model
    ports:
      - "5000:5000"

  database:
    image: postgres:15
    environment:
      POSTGRES_USER: admin
      POSTGRES_PASSWORD: admin123
      POSTGRES_DB: aidb
    volumes:
      - db_data:/var/lib/postgresql/data

  frontend:
    build: ./frontend
    ports:
      - "8081:80"
    depends_on:
      - ai_model

volumes:
  db_data:
```

Save and exit.

---

6. Verify Entire Structure

```
ls -R
```

You should see:

```
ai_model/
  app.py
  requirements.txt
  Dockerfile

frontend/
  index.html
  Dockerfile

database/
```

docker-compose.yml

---

7. Build and Run Containers

Step 7.1 — Start services

docker compose up --build

---

8. Test Application

Backend (AI Model)

Open browser:

http://localhost:5000

Expected:

{"prediction": 25.0}

Frontend

http://localhost:8081

Expected:

- Simple HTML page displayed

Activities Terminal Mar 12 4:09 PM

admin1@plcomp11: ~/ai_web_app

admin1@plcomp11: ~/ai_web_app admin1@plcomp11: ~

```
admin1@plcomp11:~$ docker --version
Docker version 28.1.1, build 4eba377
admin1@plcomp11:~$ docker compose version
Docker Compose version v5.0.1
admin1@plcomp11:~$ mkdir ai_web_app
admin1@plcomp11:~$ cd ai_web_app
admin1@plcomp11:~/ai_web_app$ mkdir ai_model frontend database
admin1@plcomp11:~/ai_web_app$ ls
ai_model  database  frontend
admin1@plcomp11:~/ai_web_app$ cd ai_model
admin1@plcomp11:~/ai_web_app/ai_model$ nano app.py
admin1@plcomp11:~/ai_web_app/ai_model$ nano requirements.txt
admin1@plcomp11:~/ai_web_app/ai_model$ nano Dockerfile
admin1@plcomp11:~/ai_web_app/ai_model$ ls
app.py  Dockerfile  requirements.txt
admin1@plcomp11:~/ai_web_app/ai_model$ cd ..
admin1@plcomp11:~/ai_web_app$ cd frontend
admin1@plcomp11:~/ai_web_app/frontend$ nano index.html
admin1@plcomp11:~/ai_web_app/frontend$ nano Dockerfile
admin1@plcomp11:~/ai_web_app/frontend$ ls
Dockerfile  index.html
admin1@plcomp11:~/ai_web_app/frontend$ cd ..
admin1@plcomp11:~/ai_web_app$ nano docker-compose.yml
admin1@plcomp11:~/ai_web_app$ ls -R
.:
ai_model  database  docker-compose.yml  frontend

./ai_model:
app.py  Dockerfile  requirements.txt

./database:

./frontend:
Dockerfile  index.html
admin1@plcomp11:~/ai_web_app$ docker compose up --build
WARN[0000] /home/admin1/ai_web_app/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] up 1/1
 ✘ Image postgres:15 Error Get "https://registry-1.docker....          15.1s
Error response from daemon: Get "https://registry-1.docker.io/v2/": net/http: request canceled while waiting for connection (Client.Timeout exceeded while awaiting headers)
admin1@plcomp11:~/ai_web_app$ docker pull postgres:15
15: Pulling from library/postgres
206356c42440: Already exists
0c725cebedaa: Pull complete
c2358c04556d: Pull complete
bae50de223a4: Pull complete
b6de334a9157: Pull complete
67266521af9d: Pull complete
fd349cbbb730: Pull complete
b480546e671b: Pull complete
8d9526eade8e: Pull complete
2f8ddeccdf2d: Pull complete
f0c010f1448f: Pull complete
f080c6a4c94c: Pull complete
```

Activities Terminal Mar 12 4:37 PM

admin1@plcomp11: ~/ai_web_app

admin1@plcomp11: ~/ai_web_app admin1@plcomp11: ~/ai_web_app

```
admin1@plcomp11:~$ mkdir ai_web_app
admin1@plcomp11:~$ cd ai_web_app
admin1@plcomp11:~/ai_web_app$
admin1@plcomp11:~/ai_web_app$ mkdir ai_model
admin1@plcomp11:~/ai_web_app$ mkdir frontend
admin1@plcomp11:~/ai_web_app$ mkdir database
admin1@plcomp11:~/ai_web_app$ cd ai_model
admin1@plcomp11:~/ai_web_app/ai_model$ nano app.py
admin1@plcomp11:~/ai_web_app/ai_model$ nano requirements.txt
admin1@plcomp11:~/ai_web_app/ai_model$ nano Dockerfile
admin1@plcomp11:~/ai_web_app/ai_model$ cd ..
admin1@plcomp11:~/ai_web_app$ cd frontend
admin1@plcomp11:~/ai_web_app/frontend$ nano index.html
admin1@plcomp11:~/ai_web_app/frontend$ nano dockerfile
admin1@plcomp11:~/ai_web_app/frontend$ cd ..
admin1@plcomp11:~/ai_web_app$ nano docker-compose.yml
admin1@plcomp11:~/ai_web_app$ docker-compose up --build
ERROR: Version in "./docker-compose.yml" is unsupported. You might be seeing this error because you're using the wrong Compose file version. Either specify a supported version (e.g "2.2" or "3.3") and pla
ce your service definitions under the `services` key, or omit the `version` key and place your service definitions at the root of the file to use version 1.
For more on the Compose file format versions, see https://docs.docker.com/compose/compose-file/
admin1@plcomp11:~/ai_web_app$ nano docker-compose.yml
admin1@plcomp11:~/ai_web_app$ nano docker-compose.yml
admin1@plcomp11:~/ai_web_app$ docker compose down
WARN[0000] /home/admin1/ai_web_app/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
admin1@plcomp11:~/ai_web_app$ docker compose up --build
WARN[0000] /home/admin1/ai_web_app/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 21.5s (20/20) FINISHED
 => [internal] load local bake definitions                                    0.0s
 => => reading from stdin 973B                                                0.0s
 => [frontend internal] load build definition from dockerfile                 0.0s
 => => transferring dockerfile: 105B                                          0.0s
 => [ai_model internal] load build definition from Dockerfile                 0.0s
 => => transferring dockerfile: 193B                                          0.0s
 => [frontend internal] load metadata for docker.io/library/nginx:alpine      2.4s
 => [ai_model internal] load metadata for docker.io/library/python:3.10-slim  1.2s
 => [ai_model internal] load .dockerignore                                    0.0s
 => => transferring context: 2B                                               0.0s
 => [ai_model 1/5] FROM docker.io/library/python:3.10-slim@sha256:4b0a8ebf16cf4563f  0.0s
 => [ai_model internal] load build context                                    0.0s
 => => transferring context: 349B                                             0.0s
 => CACHED [ai_model 2/5] WORKDIR /app                                        0.0s
 => [ai_model 3/5] COPY requirements.txt .                                    0.0s
 => [ai_model 4/5] RUN pip install --no-cache-dir -r requirements.txt        19.7s
 => [frontend internal] load .dockerignore                                    0.0s
 => => transferring context: 2B                                               0.0s
 => [frontend internal] load build context                                    0.0s
 => => transferring context: 209B                                             0.0s
 => [frontend 1/2] FROM docker.io/library/nginx:alpine@sha256:f46cb72c7df02710e693e  2.7s
 => => resolve docker.io/library/nginx:alpine@sha256:f46cb72c7df02710e693e863a983ac  0.0s
 => => sha256:f46cb72c7df02710e693e863a983ac42f6a9579058a59a35f1a 10.33kB / 10.33kB  0.0s
 => => sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa02 3.86MB / 3.86MB  0.7s
 => => sha256:593488f95c3586287709351 10ccf26bd7b3aba382b10126980e630a6d 626B / 626B  0.8s
 => => sha256:087e94b0d1e72fc0879049f5690fde107a85c327b1bcb8a7acc22f 2.50kB / 2.50kB  0.0s
```

Activities Terminal Mar 12 4:37 PM

admin1@plcomp11: ~/ai_web_app

admin1@plcomp11: ~/ai_web_app | admin1@plcomp11: ~/ai_web_app

```
ai_model-1  |  * Debug mode: off
ai_model-1  | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
ai_model-1  |  * Running on all addresses (0.0.0.0)
ai_model-1  |  * Running on http://127.0.0.1:5000
ai_model-1  |  * Running on http://172.21.0.2:5000
ai_model-1  | Press CTRL+C to quit
database-1  | performing post-bootstrap initialization ... ok
database-1  | initdb: warning: enabling "trust" authentication for local connections
database-1  | initdb: hint: You can change this by editing pg_hba.conf or using the option -A, or --auth-local and --auth-host, the next time you run initdb.
database-1  | syncing data to disk ... ok
database-1  |
database-1  |
database-1  | Success. You can now start the database server using:
database-1  |
database-1  |     pg_ctl -D /var/lib/postgresql/data -l logfile start
database-1  |
database-1  | waiting for server to start....2026-03-12 11:06:58.270 UTC [48] LOG:  starting PostgreSQL 15.17 (Debian 15.17-1.pgdg13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 6
4-bit
database-1  | 2026-03-12 11:06:58.272 UTC [48] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
database-1  | 2026-03-12 11:06:58.279 UTC [51] LOG:  database system was shut down at 2026-03-12 11:06:58 UTC
database-1  | 2026-03-12 11:06:58.282 UTC [48] LOG:  database system is ready to accept connections
database-1  |  done
database-1  | server started
database-1  | CREATE DATABASE
database-1  |
database-1  |
database-1  | /usr/local/bin/docker-entrypoint.sh: ignoring /docker-entrypoint-initdb.d/*
database-1  |
database-1  | waiting for server to shut down...2026-03-12 11:06:58.417 UTC [48] LOG:  received fast shutdown request
database-1  | .2026-03-12 11:06:58.419 UTC [48] LOG:  aborting any active transactions
database-1  | 2026-03-12 11:06:58.420 UTC [48] LOG:  background worker "logical replication launcher" (PID 54) exited with exit code 1
database-1  | 2026-03-12 11:06:58.420 UTC [49] LOG:  shutting down
database-1  | 2026-03-12 11:06:58.422 UTC [49] LOG:  checkpoint starting: shutdown immediate
database-1  | 2026-03-12 11:06:58.488 UTC [49] LOG:  checkpoint complete: wrote 922 buffers (5.6%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.010 s, sync=0.047 s, total=0.069 s; sync files=301,
longest=0.003 s, average=0.001 s; distance=4239 kB, estimate=4239 kB
database-1  | 2026-03-12 11:06:58.491 UTC [48] LOG:  database system is shut down
database-1  |  done
database-1  | server stopped
database-1  |
database-1  | PostgreSQL init process complete; ready for start up.
database-1  |
database-1  | 2026-03-12 11:06:58.531 UTC [1] LOG:  starting PostgreSQL 15.17 (Debian 15.17-1.pgdg13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
database-1  | 2026-03-12 11:06:58.531 UTC [1] LOG:  listening on IPv4 address "0.0.0.0", port 5432
database-1  | 2026-03-12 11:06:58.531 UTC [1] LOG:  listening on IPv6 address "::", port 5432
database-1  | 2026-03-12 11:06:58.535 UTC [1] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
database-1  | 2026-03-12 11:06:58.540 UTC [64] LOG:  database system was shut down at 2026-03-12 11:06:58 UTC
database-1  | 2026-03-12 11:06:58.543 UTC [1] LOG:  database system is ready to accept connections
frontend-1  | 172.21.0.1 - - [12/Mar/2026:11:07:01 +0000] "GET / HTTP/1.1" 200 170 "-" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36" "-"
frontend-1  | 2026/03/12 11:07:01 [error] 30#30: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.21.0.1, server: localhost, request: "GET /favicon.ico HTTP
/1.1", host: "localhost:8081", referrer: "http://localhost:8081/"
frontend-1  | 172.21.0.1 - - [12/Mar/2026:11:07:01 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8081/" "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.
0.0 Safari/537.36" "-"
```

Activities Docker Desktop Mar 12 4:38 PM

docker desktop PERSONAL Search Ctrl+K Sign in

A new version of Docker Desktop is available. Update

- Ask Gordon BETA
- Containers
- Images
- Volumes
- Kubernetes
- Builds
- MCP Toolkit BETA
- Docker Hub
- Docker Scout
- Extensions

Builds / frontend

frontend
kvwiioonahv034rwi3wp5ltzl

Status	Duration	Builder	Author
Completed	5.2s	default	N/A

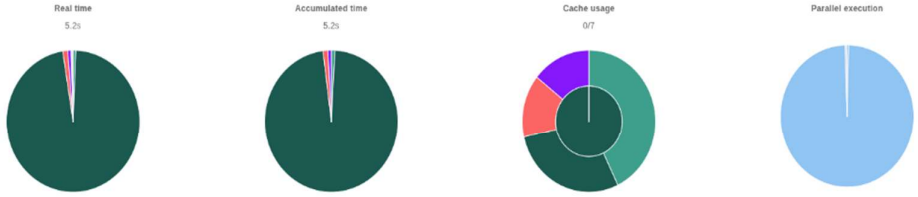
Info Source Logs History

Source details

Filter by platform: linux/amd64

File name	dockerfile
Remote source location	N/A
Revision	N/A

Build timing

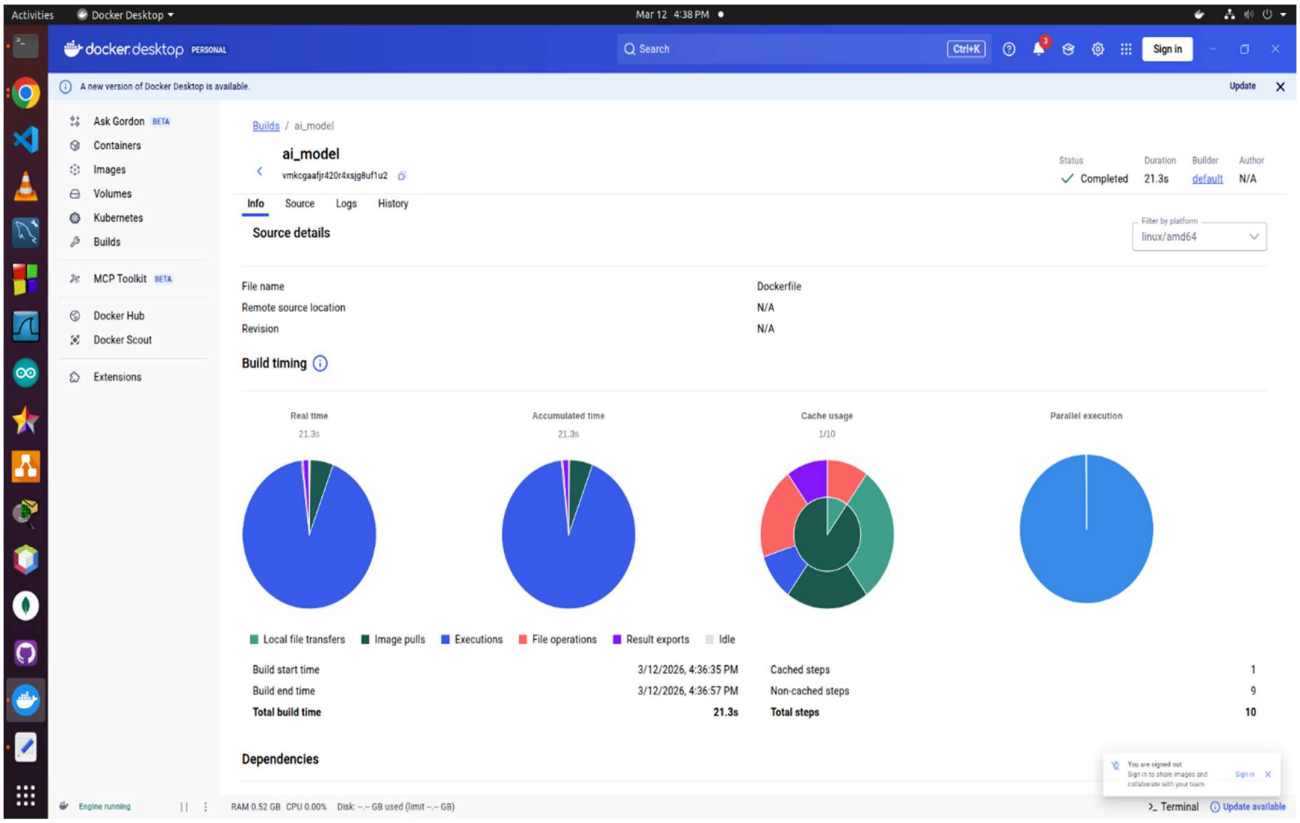


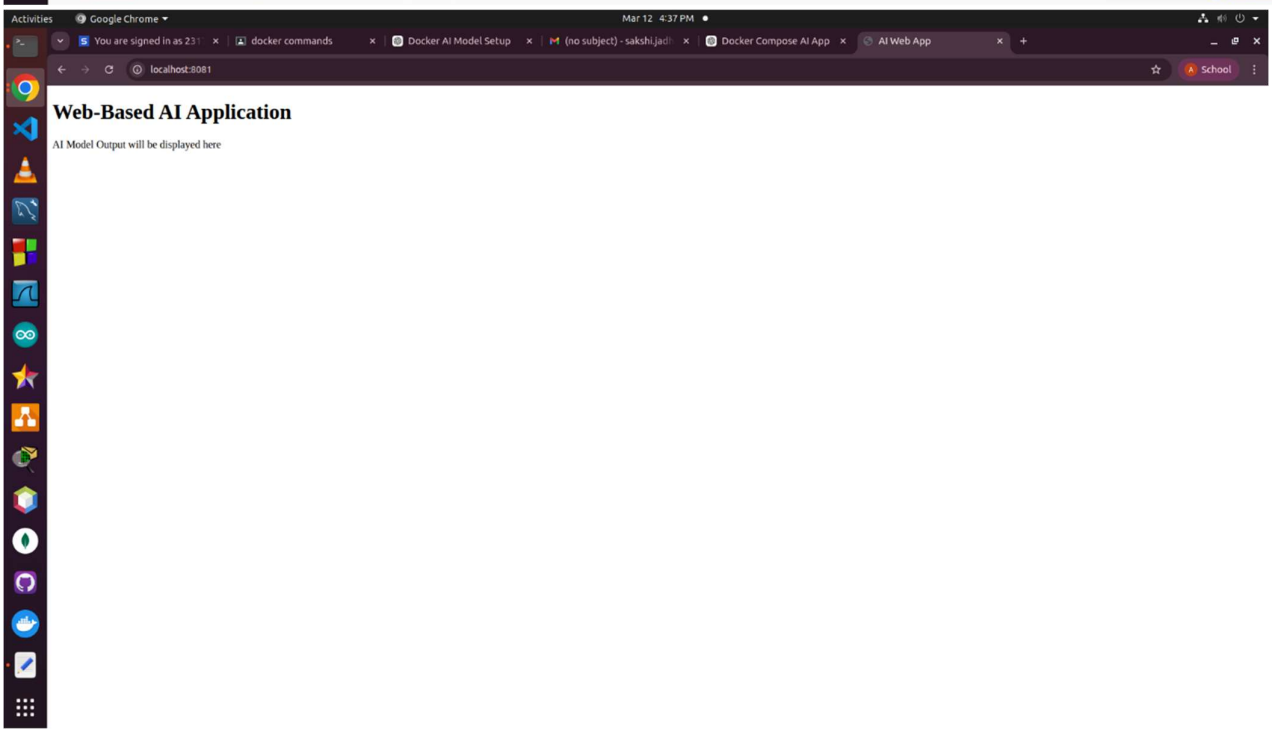
Local file transfers Image pulls File operations Result exports Idle

Build start time	3/12/2026, 4:36:35 PM	Cached steps	0
Build end time	3/12/2026, 4:36:41 PM	Non-cached steps	7
Total build time	**5.2s**	**Total steps**	**7**

Dependencies

Engine running RAM 0.52 GB CPU 0.00% Disk: --.-- GB used (limit --.-- GB) Terminal Update available


Activities
Docker Desktop
Mar 12 4:38 PM
docker desktop PERSONAL
Search
Ctrl+K
Sign in
A new version of Docker Desktop is available.
Update
Ask Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Builds / ai_model
ai_model
vmkcgaafjr420r4xsjg8uf1u2
Status
Completed
Duration
21.3s
Builder
default
Author
N/A
Info
Source
Logs
History
Source details
Filter by platform
linux/amd64
File name
Dockerfile
Remote source location
N/A
Revision
N/A
Build timing
Real time
21.3s
Accumulated time
21.3s
Cache usage
1/10
Parallel execution
Local file transfers
Image pulls
Executions
File operations
Result exports
Idle
Build start time
3/12/2026, 4:36:35 PM
Build end time
3/12/2026, 4:36:57 PM
Total build time
21.3s
Cached steps
1
Non-cached steps
9
Total steps
10
Dependencies
Engine running
RAM 0.52 GB CPU 0.00% Disk: --.-- GB used (limit --.-- GB)
Terminal
Update available


Activities
Google Chrome
Mar 12 4:37 PM
You are signed in as 231
docker commands
Docker AI Model Setup
(no subject) - sakshi.jadh
Docker Compose AI App
AI Web App
localhost:8081
School
Web-Based AI Application
AI Model Output will be displayed here